

CSCE 679 DATA VISUALIZATION · MODULE 13 · LECTURE NOTES

Claiming a Visualisation Shows Something

What a figure establishes.

MODULE OUTCOMES

State what a figure establishes.

Identify the standard overclaims.

Write a caption that makes a figure assessable.

Place this course relative to CSCE 671 and CSCE 632.

State a proportionate practice for making figures.

1 What a figure establishes

A FIGURE ESTABLISHES THAT THIS ENCODING OF THIS DERIVATION READS THIS WAY TO THIS READER

The derivation: which data was included, normalised how, aggregated to which units, and binned how — four choices made before any encoding happens (Modules 03, 08, and 11), each of which changes the picture.

The encoding: which channels carry which variables, which scales, and which axis conventions — each of which changes what can be read from the result (Modules 02 and 10).

The reading: what somebody who is not you actually extracts from the figure — which is CSCE 671 Module 01 §2's designer blindness, and is the part that is never tested.

So 'the chart shows a clear trend' is a claim about all four — and the fourth is the only one you cannot determine by looking at your own figure, because you already know what it is supposed to show (§4's step five).

The claim	The correction
'The chart shows a clear trend.'	Over which window, and with what uncertainty? (Module 09, Module 10 §2.)
'The data speaks for itself.'	It was selected, normalised, binned, and encoded by you (Module 03 §3) — see the note.
'The map shows a regional pattern.'	Normalised how? A count map is a population map (Module 08 §3).

The claim	The correction
'The clusters are clearly visible.'	In a t-SNE plot or a force-directed layout? Neither is readable that way (Module 07 §1, CSCE 676 Module 05 §4).
'This visualisation is intuitive.'	Tested with whom? Intuition is prior exposure (CSCE 671 Module 13 §2).
'The correlation is obvious from the figure.'	From the figure, or from the fitted line you drew on it? The line is a model, not data.

'The data speaks for itself' is the one to refuse — every figure is the result of a chain of choices (selection, normalisation, aggregation, binning, encoding, scaling, framing) and the phrase asserts that none of them happened. It is the most common and the emptiest claim in this subject, and refusing it is the beginning of being able to assess a figure at all.

2 The caption

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WHAT IS PLOTTED
  the variable, with its units, and the
  normalisation (Module 03 section 3)

OVER WHAT
  the population, the time window, and the spatial
  or categorical units (Modules 08, 10)

HOW IT WAS DERIVED
  the aggregation, the bin size or bandwidth, any
  smoothing, and any sampling (Modules 05, 11)

THE UNCERTAINTY
  what the intervals or bands represent, and the
  sample size (Module 09)

AND THE ENCODING, where it is unusual
  the projection for a map, the transfer function
  for a volume, the layout algorithm for a graph

WHICH IS FIVE CLAUSES, AND MOST CAPTIONS HAVE ONE.
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BECAUSE THE FIGURE CANNOT CARRY THOSE CHOICES AND THE CAPTION CAN

None of the five is visible in the figure itself — a normalisation, a bin size, a projection, a smoothing, and a sampling all leave no visual trace whatsoever, **which is exactly what makes them dangerous** rather than merely unstated.

And they each change the picture, demonstrably — **this course has shown that for every single one of them**: Module 03 §3 for normalisation, Module 05 §2 and Module 11 §2 for binning, Module 08 §1 for projection, Module 12 §3 for the transfer function.

So the caption is not a label but the figure's method section — and **a figure separated from its caption is uninterpretable, which matters because figures circulate alone**: screenshotted into a message, pasted into a slide, reposted without context.

Which argues for putting the critical statements into the figure itself — the axis label carrying the normalisation ('cases per 100,000'), an annotation naming the window, the n beside each group — because that survives being screenshotted and the caption does not. **This caption template is the course's most immediately usable artefact.**

3 The semester

Course	Its subject	Its central correction
CSCE 671	Interaction.	You cannot judge your own design, so watch somebody (its Module 01 §2).
CSCE 679 (this one)	Visual encoding.	The channel decides what can be read, and the ranking is measured (Module 02).
CSCE 632	Access.	Designing for a constraint improves the design for everybody, which is a quality argument.

All three courses replace an argument about taste with a measurement — the channel ranking, the five participants, and the assistive-technology test are the same move three times, applied to the encoding, the interaction, and the access respectively. **The replace-taste-with-measurement framing is the semester's result**, and it is what makes these three courses engineering rather than criticism.

4 A proportionate practice

- 1 · Put the main variable on position (Module 02 §1) — **which costs nothing and is the single largest improvement available to most figures**, and is the decision the ranking most clearly settles.
- 2 · Sort the categories, and fix the colour scale (Module 03 §4 and Module 04) — **both cheap, both high-return**, and both one line of code.
- 3 · Show the uncertainty and the sample size (Module 09) — **which is the honest one**, and the one that makes the figure less persuasive and more defensible.
- 4 · Write the five-clause caption (§2), and put the critical clauses into the figure itself.
- 5 · And show it to three people and ask what it says — before telling them — **which is the only test of the fourth part of §1's claim. Step 5 is the one that is never done**, and it is the only step that addresses the part of the claim you cannot assess yourself (CSCE 671 Module 01 §2).

WHERE THIS COURSE LEAVES YOU

You can specify an encoding rather than choosing a chart, and justify it from a measured ranking — **which turns a design argument into a prediction** that can be checked in an afternoon (Module 02 §4).

You can handle the hard structures — networks, hierarchies, geography, high dimension, and volumes — **and say what each representation discards**, which is the part that makes the choice defensible.

You can show uncertainty, which most published figures do not, and you know why they do not (Module 09 §1) — which makes the omission a decision rather than a habit.

The closing rule is the program's, unchanged across thirty-two courses: state what you measured, state what you assumed, and never claim more than you established. In this subject it means the five-clause caption — because a figure without its derivation is a picture, and the derivation is where the claim actually lives.

Where to see this taught differently

Original material only goes so far — hearing the same idea in another voice is how it sticks. These are free, and they cover this module specifically.

Resource	What it covers for this module
Munzner, the validation chapter (slides free)	§1's four levels — and what kind of evidence each level requires, which is the rigorous version of this module.
Hullman & Gelman — Designing for Interactive Exploratory Data Analysis (free)	§1's reading problem — and the argument that a figure encodes a model whether or not you said so.
Wilke, the telling-a-story chapters (free)	§2's captions and annotation — with the put-it-in-the-figure argument made concretely.
The IEEE VIS evaluation literature (free preprints)	§4's step five, done properly — how the field evaluates figures, and the limitations sections are the model.

Exercises

Do these in code, not on paper, unless the exercise says otherwise. The module is not finished until they run.

- 1 **Take one of your figures** and list the four parts of what it establishes.
- 2 **Find a 'the data speaks for itself' claim** and enumerate the choices behind the figure.
- 3 **Assess five published figures** against §1's table.
- 4 **Write the five-clause caption** for three of your own figures.
- 5 **Count the clauses** in ten published captions.
- 6 **Move the critical clauses into the figure**, and screenshot it to check they survived.
- 7 **Name the correction** each Semester 11 course makes.
- 8 **Run §4's five steps** on one figure, in order.
- 9 **Show a figure to three people and ask what it says**, before telling them.
- 10 **Project 2 is now due.** Submit the figure, the uncertainty representation with its justification, the baseline it beats, the five readers' open-ended descriptions, the misreadings traced to encoding decisions, and the not-shown list.

Self-check

Answer these in writing, from memory, without looking back at the notes. If you cannot, you have read this module but not yet learned it — that is useful information, not a failure.

- 1 Give the four parts of what a figure establishes.
- 2 Which part can you not assess yourself, and why?
- 3 Give six overclaims and the correction to each.
- 4 Why is 'the data speaks for itself' the one to refuse?
- 5 Give the five caption clauses.
- 6 Why must those choices go in the caption rather than the figure?
- 7 Why put the critical ones in the figure anyway?
- 8 What correction does each Semester 11 course make?
- 9 Give the five-step practice in order.
- 10 Which step is never done, and what does it uniquely address?